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Partition & covering of set-

Definition-

Let S be a given set & $A = \{A_1, A_2, \dots, A_m\}$ where each $A_i, i=1, 2, \dots, m$ is a subset of S & $\bigcup_{i=1}^m A_i = S$. Then

the set A is called a covering of S & sets $A_1, A_2, \dots, A_m$ are said to cover S .

If the elements of A , which are subsets of S , are mutually disjoint, then A is called partition of S & the sets $A_1, A_2, \dots, A_m$ are called blocks of the partition.

eg. $S = \{a, b, c\}$ & following are subsets of S .

$A = \{\{a, b\}, \{b, c\}\}$, $B = \{\{a\}, \{a, c\}\}$, $C = \{\{a\}, \{b, c\}\}$, $D = \{\{a, b, c\}\}$
 $E = \{\{a\}, \{b\}, \{c\}\}$, $F = \{\{a\}, \{a, b\}, \{a, c\}\}$

Here A & F are coverings of S . C, D, E are partitions of S .
 B is neither covering nor partition.

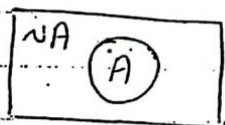
Partition $\rightarrow D$ have 1 block & E has 3 blocks. Every partition is also covering.

Two partitions are said to be equal if they are equal as sets.

Partitions of universal set-

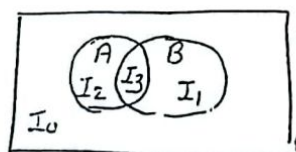
Let A is subset of E . The subsets A & $\sim A$ generate partitions of E .

$$E = A \cup \sim A$$



If A & B be any two subsets of E , then the sets
 $I_0 = \sim A \cap \sim B$, $I_1 = \sim A \cap B$, $I_2 = A \cap \sim B$, $I_3 = A \cap B$

The sets I_0, I_1, I_2 & I_3 are called complete intersections or minterms generated by subsets A & B .



I_0, I_1, I_2, I_3 are mutually disjoint.

$$E = I_0 \cup I_1 \cup I_2 \cup I_3 = \bigcup_{j=0}^3 I_j$$

The complete intersections or minterms are blocks of partitions of E generated by A & B .

Let A, B, C be three subsets of E then there are $2^3 = 8$ minterms denoted as,

$$I_0 = \sim A \cap \sim B \cap \sim C$$

$$I_1 = \sim A \cap \sim B \cap C$$

$$I_2 = \sim A \cap B \cap \sim C$$

$$I_3 = \sim A \cap B \cap C$$

$$I_4 = A \cap \sim B \cap \sim C$$

$$I_5 = A \cap \sim B \cap C$$

$$I_6 = A \cap B \cap \sim C$$

$$I_7 = A \cap B \cap C$$

If $A_1, A_2, \dots, A_n$ are n subsets of universal set E then the minterms generated by these subsets are denoted by $I_0, I_1, I_2, \dots, I_{2^n-1}$ &

$$E = \bigcup_{j=0}^{2^n-1} I_j$$

Equivalence relation -

A relation R in a set X is called an equivalence relation if it is reflexive, symmetric & transitive.

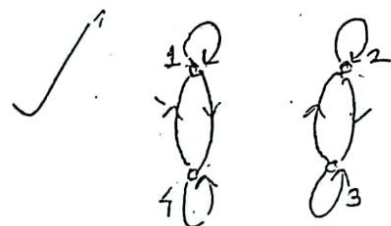
eg-1) Equality of numbers on set of real numbers.

2) Similarity of triangles on set of triangles.

3) Relation of lines being parallel on set of lines in plane

Q. 1, eg. $X = \{1, 2, 3, 4\}$ & $R = \{(1,1), (1,4), (4,1), (2,2), (2,3), (3,2), (3,3)\}$

Write its matrix & draw graph & find whether it is equivalence relation

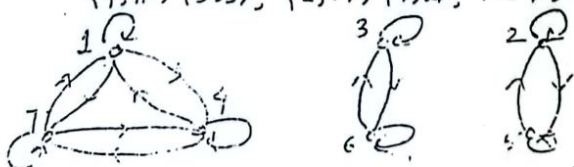


$$R = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \\ 4 & 1 & 0 & 1 \end{bmatrix}$$

It is reflexive, transitive & symmetric so it is equivalence.

Q 2 Let $X = \{1, 2, 3, \dots, 7\}$ & $R = \{(x,y) | x-y \text{ is divisible by } 3\}$. Show that R is equivalence relation. Draw graph of R .

$$R = \{(1,1), (1,4), (4,1), (2,2), (3,3), (2,5), (5,2), (3,6), (6,3), (4,7), (7,4), (4,4), (5,5), (1,7), (7,1), (6,6), (7,7)\}$$



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Congruence relation-

Let I denote set of all positive integers & let m be positive integer. For $x, y \in I$, then R is defined as

$$R = \{ \langle x, y \rangle \mid x - y \text{ is divisible by } m \}$$

Then R is called congruence modulo m relation & denoted by $\equiv$.

xRy is written as $x \equiv y \pmod{m}$.

9. Show that congruence modulo 5 is equivalence relation.

$$R = \{ \langle x, y \rangle \mid x - y \text{ is divisible by } 5 \}$$

To show R is equivalence we have to show R is reflexive, symmetric & equivalence. transitive.

→ 1) As $x - x$ is divisible by 5 so xRx i.e. it is reflexive.

2) If xRy then $x - y$ is divisible by 5

$$\text{i.e. } x - y = 5k$$

$$\text{i.e. } y - x = -5k \text{ which is divisible by } 5 \therefore yRx$$

$xRy \Rightarrow yRx$ so it is symmetric.

3) If xRy & yRz then

$$x - y = 5k$$

$$y - z = 5m$$

$$\text{i.e. } x - z = 5(k + m) \text{ i.e. } xRz$$

$\therefore xRy \wedge yRz \Rightarrow xRz \therefore R$ is transitive.

R-equivalence class-

Let R be an equivalence relation on a set X . For any $x \in X$, the set $[x]_R \subseteq X$ given by

$[x]_R = \{ y \mid y \in X \wedge xRy \}$ is called an R -equivalence class generated by $x \in X$.

$[x]_R$ is also written as x/R .

eg. If $R = \{ \langle x, y \rangle \mid (x - y) \text{ is divisible by } 3 \}$

$x, y \in \mathbb{R}$ only +ve integers $X = \{1, 2, 3, \dots, 7\}$

$$[1]_R = \{4, 7, 1\}$$

$$[4]_R = \{1, 4, 7\}$$

$$[2]_R = \{2, 5\}$$

$$[5]_R = \{5, 2\}$$

$$[3]_R = \{3, 6\}$$

$$[6]_R = \{6, 3\} \quad [7]_R = \{1, 4, 7\}$$

Properties of equivalence classes

1) For every element $x \in X$, xRx because R is reflexive
 $\therefore x \in [x]_R$

2) If xRy & $y \in X$ then $y \in [x]_R$ this implies yRx & $x \in [y]_R$ because R is symmetric.

If there is any element $z \in [y]_R$ then z must be in $[x]_R$ because $yRz \wedge xRy \Rightarrow xRz$

$$\therefore [y]_R \subseteq [x]_R \text{ & } [x]_R \subseteq [y]_R$$

$$\therefore [x]_R = [y]_R$$

$$[1]_R \subseteq [4]_R \subseteq [7]_R \text{ & } [7]_R \subseteq [4]_R \subseteq [1]_R$$

$$\therefore [1]_R = [4]_R = [7]_R$$

3) If $x \not R y$ then $[x]_R \cap [y]_R = \emptyset$

Theorem -

Every equivalence relation on set generates a unique partition of set. The blocks of partition correspond to R -equivalence classes.

eg. Let Z be set of integers & let R be relation called "congruence modulo 3" defined by

$$R = \{ \langle x, y \rangle \mid x \in \mathbb{Z} \wedge y \in \mathbb{Z} \wedge (x-y) \text{ is divisible by } 3 \}$$

Determine equivalence classes generated by z .

$$\rightarrow [0]_R = \{ \dots, -9, -6, -3, 0, 3, 6, 9, \dots \}$$

$$[1]_R = \{ \dots, -5, -2, 1, 4, 7, \dots \}$$

$$[2]_R = \{ \dots, -4, -1, 2, 5, 8, \dots \}$$

$$Z/R = \{ [0]_R, [1]_R, [2]_R \}$$

Compatibility Relation -

A relation R in X is said to be a compatibility relation if it is reflexive & symmetric. Compatibility Relation is denoted by $\approx$.

eg. Let $X = \{ \text{ball, bed, dog, let, egg} \}$ & relation R is defined as,

$$R = \{ \langle x, y \rangle \mid x, y \in X \wedge xRy \text{ if } x \text{ & } y \text{ contain some common letter} \}$$

ball $\approx$ ball, bed $\approx$ bed, dog $\approx$ dog, let $\approx$ let & egg $\approx$ egg so

R is reflexive

if ball $\approx$ bed then bed $\approx$ ball.

ball $\approx$ let then let $\approx$ ball

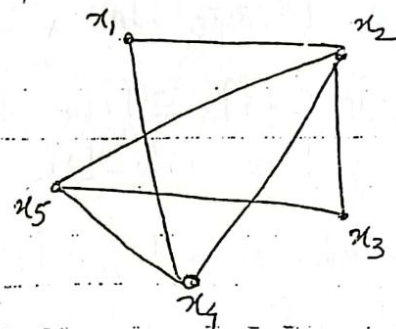
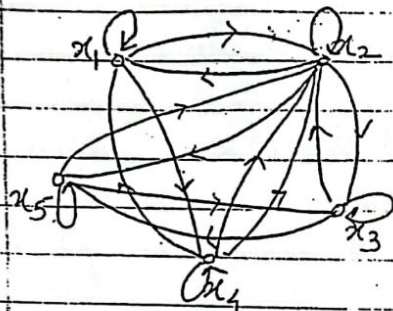
let $\approx$ egg then egg $\approx$ let.

dog $\approx$ egg then egg $\approx$ dog so it is symmetric.

But if ball $\approx$ bed $\wedge$ bed $\approx$ egg then ball $\not\approx$ egg so it is not transitive.

So R is a compatibility relation but not equivalence.
Every equivalence relation is compatibility relation.

Graph -



Maximal compatibility block-

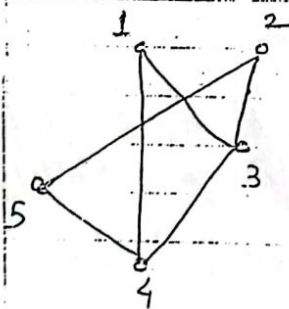
let X be a set & $\approx$ be a compatibility relation on X .

A subset $A \subseteq X$ is called maximal compatibility block if any element of A is compatible with every other element of A & no element of $X-A$ is compatible to all elements of A .

The $\{x_1, x_2, x_4\}$, $\{x_2, x_3, x_5\}$ & $\{x_2, x_4, x_5\}$ are maximal compatibility blocks.

Relation matrices

	x_2	1			
x_3	0	1			
x_4	1	1	0		
x_5	0	1	1	1	
	x_1	x_2	x_3	x_4	x_5



	1	2	3	4	5
2	0				
3	1	1			
4	1	0	1		
5	0	1	0	1	

maximal compatibility blocks-

$\{1, 3, 4\}$ $\{2, 5\}$ $\{2, 3\}$ $\{4, 5\}$